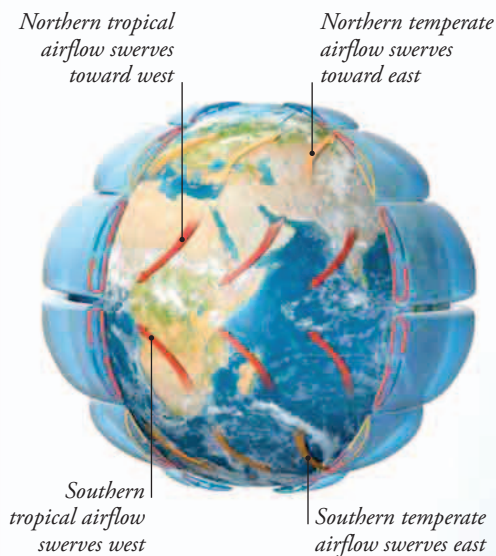




CIRCULATING AIR

In the tropics, warm air rises and flows north or south, then cools, sinks, and flows back toward the equator at low level. In temperate regions, air rises and flows toward the equator before sinking and flowing away again. At the poles, cool air sinks and flows toward the warmer temperate zones.



SPIN AND SWERVE

Earth's spin makes moving air veer off course. North of the equator the airflow swerves right, while south of the equator it swerves left. As a result, low-level air flowing toward the equator in the tropics swerves west, while air flowing away from the equator in the temperate zone swerves east. This creates the prevailing winds that blow over oceans.

Oceanic winds

As the Sun warms the atmosphere, it creates global air currents that are swept east or west by the way Earth spins in space. The air flows over the oceans in a broadly predictable pattern, creating oceanic winds that usually blow from one direction. Known as prevailing winds, they include the tropical trade winds and the stronger westerlies of cooler oceans.

TRADE WINDS

The spin and swerve effect makes the prevailing winds blow roughly from east to west over tropical oceans near the equator. They are called the trade winds because, before the invention of steamships, the tall ships that traded between the continents used them to sail west across the oceans. Trade winds are mostly gentle breezes rather than strong winds.

